📘 PRODUCT REQUIREMENTS DOCUMENT (PRD)

🎮 Product Name:

FunStack 🎯

🏷️ Tagline:

“Stacked with Fun, Powered by Full Stack”

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1. 📌 Overview

💡 Product Description:

FunStack is a full-stack web application that provides a collection of interactive mini-games and utility tools within a single platform. The application focuses on enhancing user engagement through gamification, modern UI design, and lightweight 3D animations.

🎯 Objective:

- To build a **multi-feature full-stack web application**
- To demonstrate **frontend + backend integration**
- To create an **interactive and visually engaging user experience**

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2. 👥 Target Users

- Students (primary)
- Casual users looking for quick games
- Groups of friends (multiplayer mode)

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3. 🧩 Core Features

🎯 3.1 Number Guessing Game

- Random number generation

- “Too High / Too Low” hints
- Timer mode
- Multiplayer (User1 vs User2)

🎡 3.2 Random Picker / Spin Wheel

- Input custom list
- Spin animation
- Random selection
- Multiplayer usage (decision making)

🎲 3.3 Dice Battle Game

- User vs Computer
- User vs User
- Score tracking (optional)

🧠 3.4 Quick Quiz Game

- Multiple-choice questions
- Score calculation
- Timer per question (optional)
- Multiplayer comparison

😂 3.5 Random Fun Task Generator

- Generates random tasks
- Category-based selection (Fun, Fitness, Study)
- Multiplayer mode (assign tasks)

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4. 🎮 Game Modes

👤 **Single Player:**

- User vs Computer
- Quick interaction

👥 **Multiplayer (Local):**

- User1 vs User2 (same device)
- Turn-based gameplay

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5. 🎨 UI/UX Requirements

🎯 **Design Goals:**

- Interactive and engaging UI
- Dark theme with neon accents
- Smooth animations and transitions

✨ **UI Features:**

- Dashboard with game cards
- Hover effects (scale, glow)
- 3D animations:
    - Card tilt
    - Dice rotation
    - Flip animations
- Responsive layout

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6. ⚙️ Functional Requirements

👤 **Authentication:**

- User Registration
- User Login

🎮 **Game System:**

- Game selection from dashboard
- Mode selection (single/multiplayer)
- Real-time feedback (frontend)

💾 **Data Storage:**

- Store:
  - Game results
  - Quiz scores
  - Random picker history

🌙 **Theme:**

- Dark mode toggle
- Persistent theme preference

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7. 🧠 Non-Functional Requirements

- Fast response time (<2 seconds)
- Smooth UI animations
- User-friendly interface
- Cross-browser compatibility

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8. 🛠️ Tech Stack

🎨 **Frontend:**

- HTML, CSS, JavaScript
- React.js

⚙️ **Backend:**

- Java
- Spring Boot

🗄️ **Database:**

- MySQL (recommended)

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9. 🧱 System Architecture

🔄 **Flow:**

1. User interacts with React UI
2. API calls sent to Spring Boot
3. Backend processes logic / stores data
4. Response sent back to frontend

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10. 📊 Data Model (Simplified)

Users:

- userId
- username
- password

Game Results:

- gameId
- userId
- gameType
- score / result
- timestamp

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11. 🚀 Development Phases

Phase 1:

- Setup frontend & backend
- Build dashboard UI

Phase 2:

- Implement Number Guessing Game
- Add Timer

Phase 3:

- Implement Spin Wheel
- Implement Dice Game

Phase 4:

- Implement Quiz Game
- Implement Task Generator

Phase 5:

- Add multiplayer mode
- Add backend integration

Phase 6:

- Add 3D UI effects
- Testing & polishing

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12. ⚠️ Constraints

- Limited tech stack (React + Spring Boot)
- No real-time multiplayer (no sockets)
- Focus on simplicity and performance

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13. 🔮 Future Enhancements

- Online multiplayer
- Global leaderboard
- AI-based quiz generation
- Mobile app version

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14. 🎓 Success Criteria

- All 5 games functional
- Smooth UI with animations
- Backend integration working
- Clean and responsive design
- Successful demo during evaluation

✅ FINAL SUMMARY

FunStack 🎯 is a modular, interactive full-stack web application that combines entertainment and utility through multiple mini-games, enhanced by modern UI/UX and lightweight 3D interactions.